RAJALAKSHMI ENGINEERING COLLEGE

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P2

Started on	Thursday, 20 August 2026, 11:53 AM
State	Finished
Completed on	Thursday, 20 August 2026, 12:01 PM
Time taken	8 mins 15 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:
The first line contains the no of elements in the list-n
The next n lines contain the elements.

Output:
Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

Answer:

```c
#include<stdio.h>
void swap(int * a,int *b){
    int temp =*a;
    *a=*b;
    *b=temp;
}
int partition(int arr[] , int low, int high){
    int pivot = arr[high];
    int i =(low-1);
    for(int j=low;j<high;j++){
        if(arr[j]<pivot){
            i++;
            swap(&arr[i],&arr[j]);
        }
    }
    swap(&arr[i+1],&arr[high]);
    return (i+1);
}
void quickSort(int arr[],int low,int high){
    if(low<high){
        int pi = partition(arr,low,high);
        quickSort(arr,low,pi-1);
        quickSort(arr,pi+1,high);
    }
}
int main(){
    int n;
    if(scanf("%d",&n)!=1||n<=0)
    return 0;
    int arr[n];
    for(int i =0;i<n;i++){
        scanf("%d",&arr[i]);
    }
    quickSort(arr,0,n-1);
    for(int i=0;i<n;i++){
        printf("%d%c",arr[i],(i ==n-1)? '\n':' ');
    }
    return 0;
}
```

	Input	Expected	Got	
✔	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✔
✔	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✔

	Input	Expected	Got	
✔	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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